CORE PURE / COMPLEX NUMBERS

De Moivre's Theorem

Mark scheme 18 questions · 184 marks

Questions

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	Total	**184**	

Reading the mark scheme

These rules apply throughout, unless a note says otherwise.

Marks

Every mark is worth one, so **M1 A1** on a row is two marks. Only marks coded ft follow through.

Code	Meaning
M1	Method: a correct method, applied to this question. A slip doesn't cost it if the method is clear. A formula must be used, not just quoted.
dM1	Dependent method: as M1, but only if the method mark before it was earned, unless its note names another.
A1	Accuracy: a correct answer or step. It needs the method mark it follows.
A1ft	As A1, following through from an earlier answer that the note names.
A1 AG	The last mark of a part whose answer is printed in the question: complete working, with no errors.
B1	A correct value, fact or statement. It needs no method mark.
B1ft	As B1, following through from an earlier answer that the note names.
E1	A written explanation, interpretation or conclusion. Concluding a proof by induction needs every earlier mark.

Abbreviations

ft and AG qualify a code in the Marks column. The others appear in the notes.

ft	follow through	**AG**	answer given in the question
cao	correct answer only, with no follow-through	**cso**	correct solution only: no errors in the part
oe	or an equivalent form	**awrt**	any answer that rounds to this value
soi	seen, or implied by later working	**isw**	ignore working after a correct answer
SC	special case: credit outside the main method	**their**	the student's own earlier value
condone	a slip that doesn't cost the mark	[ ]	working the mark doesn't need
s.f.	significant figures	**d.p.**	decimal places

Reading the scheme

- Each row is a line of working, with the marks it earns beside it. Where a row carries two marks, each note starts with its code.
- [$\boldsymbol{n}$] ends each part with its total.
- A shaded *Alternative* is a second method for the same total. Mark an answer by the method it follows.
- Any other valid method earns the equivalent marks, unless the question names a method or says *hence*.
- A sketch earns one B1 for each feature named below its figure.

Accuracy and answers

- Unless a question says otherwise, give answers to 3 significant figures, and angles in degrees to 1 decimal place. Angles in radians take 3 significant figures.
- A final answer with more figures is accepted if it rounds to the scheme's value; one with fewer is not. An answer spoiled by rounding too early loses its accuracy mark.
- Where a question asks for an exact answer, a decimal loses the final accuracy mark. Otherwise either is accepted.
- Wrong or missing units cost nothing unless the question asks for them.
- An omitted $\mathrm{d}\theta$, $\mathrm{d}x$ or $\mathrm{d}u$ in an integral is not penalised where the variable of integration is clear from the integrand and the limits. It is penalised where it is genuinely ambiguous: a substitution part-way through, two variables in play, or limits in the wrong variable.
- A correct answer with no working earns full marks, unless the question needs the working: show that, prove, hence, a method it names, or an exact answer. A wrong answer with no working earns nothing. Calculator answers are treated the same way.
- A correct answer, once reached, isn't spoiled by later working (isw).

Value of $\boldsymbol{g}$

- Take $g = 9.8\,\mathrm{m\,s^{-2}}$ unless a question states a value, which is then the only one accepted. Answers that use $g = 9.8$ may be given to 2 or 3 significant figures.
- Where a question doesn't state g, an answer worked with $g = 10$ throughout earns the same marks, keeping any value printed in the question. The worked solutions list the final answers with $g = 10$.

Question	Scheme	Marks	Notes
1	$z - z^{-1} = 2i\sin\theta,\ z + z^{-1} = 2\cos\theta$ $\Rightarrow \sin\theta\cos^4\theta = \frac{1}{32i}(z - z^{-1})(z + z^{-1})^4$	**B1**	Both, stated or used.
	$(z - z^{-1})(z^4 + 4z^2 + 6 + 4z^{-2} + z^{-4})$	**M1**	Multiplies out $(z - z^{-1})(z + z^{-1})^4$ to a sum of powers of z, in any order of multiplication. Condone coefficient and sign slips.
	$= z^5 + 3z^3 + 2z - 2z^{-1} - 3z^{-3} - z^{-5}$	**A1**	Correct expansion, oe; the factor $\frac{1}{32i}$ need not be shown here.
	$(z^5 - z^{-5}) + 3(z^3 - z^{-3}) + 2(z - z^{-1})$ $= 2i\sin 5\theta + 6i\sin 3\theta + 4i\sin\theta$	**M1**	Pairs terms as $z^n - z^{-n}$ and replaces each by $2i\sin n\theta$, keeping their coefficients.
	$A = \frac{1}{8},\ B = \frac{3}{16},\ C = \frac{1}{16}$	**A1**	cso; withhold if an i or the factor $\frac{1}{32}$ is lost and recovered. Accept the identity written with these values.
	Total for Question 1	**5**	

Question	Scheme	Marks	Notes
2(a)	$z^n = \cos n\theta + i\sin n\theta$ $z^{-n} = \cos(-n\theta) + i\sin(-n\theta)$ $= \cos n\theta - i\sin n\theta$	**M1**	Uses de Moivre's theorem to write both z^n and z^{-n} in terms of $\cos n\theta$ and $\sin n\theta$; z^{-n} may come from $\frac{1}{z^n}$ by realising the denominator.
	$z^n + \frac{1}{z^n} = 2\cos n\theta$	**A1**	From correct expressions for both powers, with $\cos(-n\theta)$ and $\sin(-n\theta)$ correctly simplified.
	$z^n - \frac{1}{z^n} = (\cos n\theta + i\sin n\theta) - (\cos n\theta - i\sin n\theta)$ $= 2i\sin n\theta$	**A1 AG**	Correct subtraction of the two expressions leading without error to the printed result.
		[3]	
2(b)	$\left(z + \frac{1}{z}\right)^3 = z^3 + \frac{1}{z^3} + 3\left(z + \frac{1}{z}\right)$ $\Rightarrow 8\cos^3\theta = 2\cos 3\theta + 6\cos\theta$ $\left(z - \frac{1}{z}\right)^3 = z^3 - \frac{1}{z^3} - 3\left(z - \frac{1}{z}\right)$ $\Rightarrow -8i\sin^3\theta = 2i\sin 3\theta - 6i\sin\theta$	**M1**	Expands both cubes binomially, groups into $z^3 \pm z^{-3}$ and $z \pm z^{-1}$, and replaces each by (a). Condone sign and coefficient slips.
	$\cos 3\theta = 4\cos^3\theta - 3\cos\theta$ $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$	**A1**	Both correct, from working with $(2i\sin\theta)^3 = -8i\sin^3\theta$ correct.
	$\tan 3\theta = \frac{3\sin\theta - 4\sin^3\theta}{4\cos^3\theta - 3\cos\theta}$ $= \frac{3\tan\theta\sec^2\theta - 4\tan^3\theta}{4 - 3\sec^2\theta}$	**M1**	Writes $\tan 3\theta$ as their $\sin 3\theta$ over their $\cos 3\theta$, each a cubic in $\sin\theta$ or $\cos\theta$, and divides through by a power of $\cos\theta$ to introduce $\tan\theta$.
	$= \frac{3\tan\theta(1+\tan^2\theta) - 4\tan^3\theta}{4 - 3(1+\tan^2\theta)}$ $= \frac{3\tan\theta - \tan^3\theta}{1 - 3\tan^2\theta}$	**A1 AG**	From correct expressions for $\sin 3\theta$ and $\cos 3\theta$, with $\sec^2\theta = 1 + \tan^2\theta$ oe used correctly, leading without error to the printed result.
		[4]	
	Total for Question 2	**7**	

Question	Scheme	Marks	Notes
3	$\cos 6\theta + i\sin 6\theta = (\cos\theta + i\sin\theta)^6$ $= \cos^6\theta\,(1+it)^6$	**M1**	Applies de Moivre's theorem and equates $\cos 6\theta + i\sin 6\theta$ to $\cos^6\theta\,(1+it)^6$.
	$(1+it)^6 = 1 + 6(it) + 15(it)^2 + 20(it)^3$ $+ 15(it)^4 + 6(it)^5 + (it)^6$	**M1**	Expands with seven terms in powers of it and binomial coefficients; condone one coefficient slip.
	$= (1 - 15t^2 + 15t^4 - t^6)$ $+ i(6t - 20t^3 + 6t^5)$	**A1**	Correct, with powers of i simplified; need not be grouped into real and imaginary parts.
	$\cos 6\theta = \cos^6\theta\,(1 - 15t^2 + 15t^4 - t^6)$ $\sin 6\theta = \cos^6\theta\,(6t - 20t^3 + 6t^5)$	**M1 A1**	**M1** Equates real parts and imaginary parts of their two expressions for $(\cos\theta + i\sin\theta)^6$. **A1** Both correct, including the factor $\cos^6\theta$.
	$\tan 6\theta = \frac{\sin 6\theta}{\cos 6\theta} = \frac{\cos^6\theta\,(6t-20t^3+6t^5)}{\cos^6\theta\,(1-15t^2+15t^4-t^6)}$	**M1**	Divides their $\sin 6\theta$ by their $\cos 6\theta$, or the imaginary part by the real part, and cancels $\cos^6\theta$.
	$\tan 6\theta = \frac{6\tan\theta - 20\tan^3\theta + 6\tan^5\theta}{1 - 15\tan^2\theta + 15\tan^4\theta - \tan^6\theta}$	**A1 AG**	cso: printed result from correct working with no errors in powers of i or signs. Condone the final line left in t.
	Total for Question 3	**7**	

Question	Scheme	Marks	Notes
4(a)	$(\cos x + i\sin x)^6 = \cos 6x + i\sin 6x$ $= \cos^6 x + 6i\cos^5 x \sin x$ $+ 15i^2\cos^4 x \sin^2 x + \ldots$	**M1**	Applies de Moivre to $(\cos x + i\sin x)^6$ and expands binomially, at least the terms giving the imaginary part. Condone slips in coefficients or powers of i.
	$\sin 6x = 6\cos^5 x \sin x - 20\cos^3 x \sin^3 x$ $+ 6\cos x \sin^5 x$	**A1**	Equates imaginary parts correctly, oe.
	$\sin 6x = 2\sin x \cos x\big(3\cos^4 x$ $- 10\cos^2 x(1-\cos^2 x) + 3(1-\cos^2 x)^2\big)$	**M1**	Uses $\sin^2 x + \cos^2 x = 1$ to write the bracket in powers of $\cos x$ alone, by any route.
	$\sin 6x = \sin 2x\,(\ldots\cos^4 x + \ldots\cos^2 x + \ldots)$	**dM1**	Expands and collects to a quartic in $\cos x$, with $2\sin x\cos x$ replaced by $\sin 2x$. Condone arithmetic slips.
	$\sin 6x = \sin 2x\,(16\cos^4 x - 16\cos^2 x + 3)$ $a = 16,\ b = -16,\ c = 3$	**A1**	cso. Correct identity or a, b, c stated.
		[5]	
4(b)	$\sin 2\theta(16\cos^4\theta - 16\cos^2\theta + 3) = \sin 2\theta\cos 2\theta$ $\Rightarrow 16\cos^4\theta - 16\cos^2\theta + 3 = 2\cos^2\theta - 1$	**M1**	Uses their identity of the form $\sin 2\theta(a\cos^4\theta + b\cos^2\theta + c)$, cancels $\sin 2\theta$ and writes $\cos 2\theta$ as $2\cos^2\theta - 1$. Condone sign slips.
	$16\cos^4\theta - 18\cos^2\theta + 4 = 0$	**A1**	oe, such as $8\cos^4\theta - 9\cos^2\theta + 2 = 0$.
	$\cos^2\theta = \frac{9\pm\sqrt{17}}{16} \Rightarrow \theta = \ldots$	**M1**	Solves their three-term quadratic in $\cos^2\theta$ by any means, calculator allowed, and inverts to find at least one θ.
	$\theta = 0.438,\ 0.986$	**A1**	Both, awrt 0.438 and 0.986, and no others in $0 < \theta < \frac{\pi}{2}$.
		[4]	
	Total for Question 4	**9**	

Question	Scheme	Marks	Notes
5(a)	$(z - z^{-1})^6 = (2i\sin\theta)^6 = -64\sin^6\theta$	**B1**	
	$(z - z^{-1})^6$ $= z^6 - 6z^4 + 15z^2 - 20 + 15z^{-2} - 6z^{-4} + z^{-6}$	**M1 A1**	**M1** Expands using binomial coefficients 1, 6, 15, 20 with powers of z and z^{-1}. Condone sign and power slips; need not be simplified. **A1** All seven terms correct; powers need not be simplified.
	$(z^6 + z^{-6}) - 6(z^4 + z^{-4}) + 15(z^2 + z^{-2}) - 20$ $= 2\cos 6\theta - 12\cos 4\theta + 30\cos 2\theta - 20$	**M1**	Pairs terms and uses $z^n + z^{-n} = 2\cos n\theta$ for at least two pairs. Condone coefficient slips.
	$-64\sin^6\theta = 2\cos 6\theta - 12\cos 4\theta + 30\cos 2\theta - 20$ $\Rightarrow \sin^6\theta = \frac{1}{32}(10 - 15\cos 2\theta + 6\cos 4\theta - \cos 6\theta)$	**A1 AG**	Equates the two expressions and reaches the printed identity without error; combined rearrangement accepted.
		[5]	
5(b)	$x = \tan\theta,\ \mathrm{d}x = \sec^2\theta\,\mathrm{d}\theta$ $\Rightarrow \int \frac{\tan^6\theta}{(\sec^2\theta)^4}\sec^2\theta\,\mathrm{d}\theta$	**M1**	Substitutes fully: $\mathrm{d}x$ replaced by $\sec^2\theta\,\mathrm{d}\theta$ and $1 + x^2$ by $\sec^2\theta$ or $1 + \tan^2\theta$. Condone slips in powers.
	$\int_0^{\pi/4}\sin^6\theta\,\mathrm{d}\theta$	**A1**	
	$\frac{1}{32}\int_0^{\pi/4}(10 - 15\cos 2\theta + 6\cos 4\theta - \cos 6\theta)\,\mathrm{d}\theta$ $= \frac{1}{32}\left[10\theta - \frac{15}{2}\sin 2\theta + \frac{3}{2}\sin 4\theta - \frac{1}{6}\sin 6\theta\right]_0^{\pi/4}$	**M1 A1**	**M1** Replaces $\sin^6\theta$ by the result of (a) and integrates to the form $a\theta + \sum b_k \sin k\theta$. Condone sign and coefficient slips. **A1** Correct integral; the factor $\frac{1}{32}$ may be applied later.
	$\frac{1}{32}\left(\frac{10\pi}{4} - \frac{15}{2} + 0 + \frac{1}{6}\right)$	**dM1**	Substitutes $\frac{\pi}{4}$ and 0 into their integral and subtracts; the zero from the lower limit need not be shown.
	$= \frac{15\pi - 44}{192}$	**A1**	Any exact equivalent, such as $\frac{5\pi}{64} - \frac{11}{48}$.
		[6]	
	Total for Question 5	**11**	

Question	Scheme	Marks	Notes
6(a)	$\cos 5\theta + i\sin 5\theta = (\cos\theta + i\sin\theta)^5$ $= \cos^5\theta + 5i\cos^4\theta\sin\theta - 10\cos^3\theta\sin^2\theta + \ldots$	**M1**	Uses de Moivre with a binomial expansion of $(\cos\theta + i\sin\theta)^5$; at least the real terms, with binomial coefficients. Condone slips in powers of i.
	$\cos 5\theta = \cos^5\theta - 10\cos^3\theta\sin^2\theta + 5\cos\theta\sin^4\theta$	**A1**	Equates real parts correctly.
	$= \cos^5\theta - 10\cos^3\theta(1-\cos^2\theta)$ $+ 5\cos\theta(1-\cos^2\theta)^2$	**M1**	Replaces $\sin^2\theta$ by $1-\cos^2\theta$ in both terms containing $\sin\theta$.
	$= \cos^5\theta - 10\cos^3\theta + 10\cos^5\theta$ $+ 5\cos\theta - 10\cos^3\theta + 5\cos^5\theta$	**dM1**	Expands the brackets and collects to a form $a\cos^5\theta + b\cos^3\theta + c\cos\theta$. Condone arithmetic slips.
	$\cos 5\theta = 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta$ $2\cos 5\theta = 32\cos^5\theta - 40\cos^3\theta + 10\cos\theta$	**A1 AG**	Correct real part, substituted and expanded without error to the printed identity, including the doubling; combined steps accepted.
		[5]	
6(b)	$x = 2\cos\theta \Rightarrow x^5 - 5x^3 + 5x$ $= 32\cos^5\theta - 40\cos^3\theta + 10\cos\theta = 2\cos 5\theta$	**M1**	Substitutes $x = 2\cos\theta$ and uses (a) to write $x^5 - 5x^3 + 5x$ as $2\cos 5\theta$.
	$2\cos 5\theta + 1 = 0 \Rightarrow \cos 5\theta = -\frac{1}{2}$	**A1**	
	$5\theta = \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{8\pi}{3}, \frac{10\pi}{3}, \frac{14\pi}{3} \Rightarrow \theta = \ldots$	**M1**	Finds at least two values of 5θ for their $\cos 5\theta = k$, $\lvert k\rvert \leq 1$, and divides by 5.
	$x = 2\cos\frac{2\pi}{15}, 2\cos\frac{4\pi}{15}, 2\cos\frac{8\pi}{15}, 2\cos\frac{2\pi}{3}, 2\cos\frac{14\pi}{15}$	**A1**	All five, each as $2\cos q\pi$ oe (e.g. $2\cos\frac{16\pi}{15}$), no repeated root. Decimals, -1, or θ values alone score A0.
		[4]	
	Total for Question 6	**9**	

Question	Scheme	Marks	Notes
7(a)	$z^n = \cos n\theta + i\sin n\theta$ $z^{-n} = \cos(-n\theta) + i\sin(-n\theta)$ $= \cos n\theta - i\sin n\theta$	**M1**	Applies de Moivre to both z^n and z^{-n}, or finds $\frac{1}{z^n}$ from z^n using the conjugate. Simplification of $\cos(-n\theta)$, $\sin(-n\theta)$ not needed.
	$z^n + \frac{1}{z^n} = 2\cos n\theta$	**A1**	From a correctly simplified $\frac{1}{z^n}$.
	$z^n - \frac{1}{z^n} = 2i\sin n\theta$	**A1 AG**	Correct subtraction leading without error to the printed result.
		[3]	
7(b)	$(2\cos\theta)^3(2i\sin\theta)^2 = -32\sin^2\theta\cos^3\theta$	**B1**	
	$\left(z^3 + 3z + \frac{3}{z} + \frac{1}{z^3}\right)\left(z^2 - 2 + \frac{1}{z^2}\right) = \ldots$	**M1**	Multiplies out the product, by binomial expansions or otherwise, to a sum of powers of z. Condone coefficient slips.
	$z^5 + z^3 - 2z - \frac{2}{z} + \frac{1}{z^3} + \frac{1}{z^5}$	**A1**	
	$\left(z^5 + \frac{1}{z^5}\right) + \left(z^3 + \frac{1}{z^3}\right) - 2\left(z + \frac{1}{z}\right)$ $= 2\cos 5\theta + 2\cos 3\theta - 4\cos\theta$	**M1**	Pairs z^k with z^{-k} in their expansion and replaces each pair by $2\cos k\theta$.
	$A = -\frac{1}{16},\ B = -\frac{1}{16},\ C = \frac{1}{8}$	**A1**	cso, from equating with $-32\sin^2\theta\cos^3\theta$. Accept the identity written with these values.
		[5]	
	Total for Question 7	**8**	

Question	Scheme	Marks	Notes
8(a)	$(1+i\tan\theta)^4 = \sec^4\theta(\cos\theta+i\sin\theta)^4$	**B1**	The given form raised to the fourth power, $\sec^4\theta(\cos\theta+i\sin\theta)^4$ oe. Not for the given form alone.
	$= \sec^4\theta(\cos 4\theta + i\sin 4\theta)$	**M1**	Applies de Moivre's theorem to $(\cos\theta+i\sin\theta)^4$. Condone a slip in the power of $\sec\theta$.
	$= \frac{\cos 4\theta + i\sin 4\theta}{\cos^4\theta}$	**A1 AG**	Printed result reached without error from the fourth power of the given form and de Moivre's theorem.
		[3]	
8(b)	$(1+i\tan\theta)^4 = 1 + 4i\tan\theta + 6i^2\tan^2\theta + 4i^3\tan^3\theta + i^4\tan^4\theta$	**M1**	Expands to five terms with coefficients 1, 4, 6, 4, 1 and powers of $i\tan\theta$. Powers of i need not be simplified.
	$= (1 - 6\tan^2\theta + \tan^4\theta) + i\,4\tan\theta(1-\tan^2\theta)$	**A1**	Correct real and imaginary parts, in any arrangement.
	Equates parts with (a): $\frac{\cos 4\theta}{\cos^4\theta} = 1 - 6\tan^2\theta + \tan^4\theta$ $\cos^4\theta = \frac{1}{(1+\tan^2\theta)^2}$	**M1**	Equates real or imaginary parts with (a) and replaces $\cos^4\theta$ using $\sec^2\theta = 1+\tan^2\theta$. Condone slips in their expansion.
	$\cos 4\theta = \frac{1-6\tan^2\theta+\tan^4\theta}{(1+\tan^2\theta)^2}$ $\sin 4\theta = \frac{4\tan\theta(1-\tan^2\theta)}{(1+\tan^2\theta)^2}$	**A1 AG**	Both printed results, reached without error from correct real and imaginary parts.
		[4]	
8(c)	$\tan 4\theta = \frac{\sin 4\theta}{\cos 4\theta}$ $= \frac{4\tan\theta(1-\tan^2\theta)}{(1+\tan^2\theta)^2} \div \frac{1-6\tan^2\theta+\tan^4\theta}{(1+\tan^2\theta)^2}$	**M1**	Divides their $\sin 4\theta$ by their $\cos 4\theta$ from (b), or the imaginary part of the expansion by the real part.
	$\tan 4\theta = \frac{4\tan\theta(1-\tan^2\theta)}{1-6\tan^2\theta+\tan^4\theta}$	**A1 AG**	Printed result from the correct quotient with no errors.
		[2]	
	Total for Question 8	**9**	

Question	Scheme	Marks	Notes
9(a)	$\cos 6\theta + i\sin 6\theta = (c+is)^6$ $= c^6 + 6c^5(is) + 15c^4(is)^2 + \ldots + (is)^6$	**M1**	Equates $\cos 6\theta + i\sin 6\theta$ to a binomial expansion of $(\cos\theta + i\sin\theta)^6$ with at least the real terms. Condone slips in coefficients or powers of i.
	$\cos 6\theta = c^6 - 15c^4s^2 + 15c^2s^4 - s^6$	**A1**	
	$\cos 6\theta = c^6 - 15c^4(1-c^2) + 15c^2(1-c^2)^2$ $-(1-c^2)^3 = \ldots$	**M1**	Replaces every $\sin^2\theta$ by $1-\cos^2\theta$ and expands to a polynomial in $\cos\theta$. Condone expansion slips.
	$\cos 6\theta = 32c^6 - 48c^4 + 18c^2 - 1$	**A1**	
	$\sec 6\theta = \dfrac{1}{\text{“}32c^6 - 48c^4 + 18c^2 - 1\text{”}} \times \dfrac{\sec^6\theta}{\sec^6\theta}$	**M1**	Takes the reciprocal of their polynomial in $\cos\theta$ and multiplies numerator and denominator by $\sec^6\theta$, or writes each $\cos^n\theta$ as $\sec^{-n}\theta$.
	$\sec 6\theta = \dfrac{\sec^6\theta}{32 - 48\sec^2\theta + 18\sec^4\theta - \sec^6\theta}$	**A1 AG**	From the correct polynomial, reaches the printed result without error; a one-step multiplication is accepted.
		[6]	
9(b)	$x = \sec\theta$: $\sec^6\theta - 18\sec^4\theta + 48\sec^2\theta - 32 = 0$ $\Rightarrow 32 - 48\sec^2\theta + 18\sec^4\theta - \sec^6\theta = 0$	**M1**	Substitutes $x = \sec\theta$ and identifies the equation as the denominator in (a) equal to zero, sign change allowed.
	$\sec 6\theta$ undefined $\Rightarrow \cos 6\theta = 0$	**A1**	oe such as $\sec 6\theta$ undefined or infinite.
	$6\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \ldots \Rightarrow \theta = \ldots$	**M1**	Solves their $\cos 6\theta = 0$ for at least two values of θ giving distinct values of $\sec\theta$.
	$x = \sec\frac{\pi}{12}, \sec\frac{\pi}{4}, \sec\frac{5\pi}{12}, \sec\frac{7\pi}{12}, \sec\frac{3\pi}{4}, \sec\frac{11\pi}{12}$	**A1**	Six distinct roots in the form $\sec q\pi$, no others; any equivalent q. Decimals, or $\pm\sqrt{2}$, in place of $\sec q\pi$ score A0.
		[4]	
	Total for Question 9	**10**	

Question	Scheme	Marks	Notes
10(a)	$\cos 5\theta + i\sin 5\theta = (\cos\theta + i\sin\theta)^5$ $= \cos^5\theta + 5i\cos^4\theta\sin\theta - 10\cos^3\theta\sin^2\theta$ $- 10i\cos^2\theta\sin^3\theta + 5\cos\theta\sin^4\theta + i\sin^5\theta$	**M1**	Equates $\cos 5\theta + i\sin 5\theta$ to $(\cos\theta + i\sin\theta)^5$ and attempts a binomial expansion with at least the three real terms. Condone coefficient or sign slips.
	$\cos 5\theta = \cos^5\theta - 10\cos^3\theta\sin^2\theta + 5\cos\theta\sin^4\theta$	**A1**	Correct real part; imaginary terms need not be seen.
	$\cos 5\theta = \cos^5\theta - 10\cos^3\theta(1-\cos^2\theta)$ $+ 5\cos\theta(1-\cos^2\theta)^2$	**M1**	Substitutes $\sin^2\theta = 1 - \cos^2\theta$ into their real part to give an expression in $\cos\theta$ only.
	$= \cos^5\theta - 10\cos^3\theta + 10\cos^5\theta$ $+ 5\cos\theta - 10\cos^3\theta + 5\cos^5\theta$	**dM1**	Expands the brackets, including $(1-\cos^2\theta)^2$, and collects like powers of $\cos\theta$. Condone slips.
	$\cos 5\theta = 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta$	**A1 AG**	Correct real part leading without error to the printed result with $\cos 5\theta$ as LHS; combined steps accepted.
		[5]	
10(b)	$\cos 5\theta = \cos 2\theta$	**B1**	Replaces the LHS by $\cos 5\theta$ using (a).
	$5\theta = 2n\pi \pm 2\theta \Rightarrow \theta = \ldots$	**M1**	Solves by a general solution of $\cos A = \cos B$ with both signs, a factor formula, or listing equal-cosine angles over a sufficient range, leading to $\theta = \ldots$.
	$\theta = \frac{2n\pi}{3}$	**A1**	Or its values 0 and $\frac{2\pi}{3}$ in range.
	$\theta = \frac{2n\pi}{7}$	**A1**	Or its values 0, $\frac{2\pi}{7}$, $\frac{4\pi}{7}$, $\frac{6\pi}{7}$ in range.
	$\theta = 0,\ \frac{2\pi}{7},\ \frac{4\pi}{7},\ \frac{2\pi}{3},\ \frac{6\pi}{7}$	**A1**	All five exact values, including 0, and no others in $0 \le \theta \le \pi$.
		[5]	
Total for Question 10		**10**	

Question	Scheme	Marks	Notes
11(a)	$(\cos\theta + i\sin\theta)^6 = \cos 6\theta + i\sin 6\theta$ $= \cos^6\theta + 6i\cos^5\theta\sin\theta$ $- 15\cos^4\theta\sin^2\theta + \ldots$	**M1**	Equates $\cos 6\theta + i\sin 6\theta$ to a binomial expansion of $(\cos\theta + i\sin\theta)^6$, at least its real terms, with powers of i applied. Condone coefficient and sign slips.
	$\cos 6\theta = \cos^6\theta - 15\cos^4\theta\sin^2\theta$ $+ 15\cos^2\theta\sin^4\theta - \sin^6\theta$	**A1**	
	$u = \sin^2\theta$: $\cos 6\theta = (1-u)^3 - 15(1-u)^2u$ $+ 15(1-u)u^2 - u^3$	**M1**	Replaces $\cos^2\theta$ by $1 - \sin^2\theta$ throughout their real part.
	$= 1 - 18u + 48u^2 - 32u^3$ $= (1-2u)(\ldots)$	**dM1**	Expands and collects into a cubic in $\sin^2\theta$, then factorises it or expands the printed product to compare. Condone slips.
	$\cos 6\theta \equiv (1 - 2\sin^2\theta)(16\sin^4\theta$ $- 16\sin^2\theta + 1)$	**A1 AG**	Correct substituted expression leading without error to the printed identity; matching the expanded printed RHS with a conclusion is accepted.
		[5]	
11(b)	$(1-2u)(16u^2 - 16u + 1) = 0 \Rightarrow u = \ldots$	**M1**	Sets each factor to zero and solves the quadratic in $\sin^2\theta$ by formula with their values or completing the square.
	$u = \frac{1}{2},\ \frac{2\pm\sqrt{3}}{4}$	**A1**	All three, oe such as $\frac{16\pm8\sqrt{3}}{32}$.
	$\cos\left(6\times\frac{\pi}{12}\right) = \cos\frac{\pi}{2} = 0$ $0 < \frac{\pi}{12} < \frac{\pi}{4} \Rightarrow \sin^2\frac{\pi}{12} < \frac{1}{2}$	**M1**	Uses $\theta = \frac{\pi}{12}$ as a solution of $\cos 6\theta = 0$ and gives a reason rejecting $\frac{1}{2}$ and $\frac{2+\sqrt{3}}{4}$, such as $\sin^2\frac{\pi}{12} < \frac{1}{2}$.
	$\sin^2\left(\frac{\pi}{12}\right) = \frac{2-\sqrt{3}}{4}$	**A1 AG**	cso, with both other values rejected.
		[4]	
11(c)	$\sin^2\left(\frac{5\pi}{12}\right) = \frac{2+\sqrt{3}}{4}$	**B1**	Any exact form, here and below.
	$\sin^2\left(\frac{7\pi}{12}\right) = \frac{2+\sqrt{3}}{4}$	**B1**	
	$\sin^2\left(\frac{11\pi}{12}\right) = \frac{2-\sqrt{3}}{4}$	**B1**	
		[3]	
	Total for Question 11	**12**	

Question	Scheme	Marks	Notes
12(i)	$\cos 5\theta + i\sin 5\theta = (\cos\theta + i\sin\theta)^5$ $= \cos^5\theta + 5i\cos^4\theta\sin\theta$ $- 10\cos^3\theta\sin^2\theta + \ldots$	**M1**	Uses de Moivre with a binomial expansion of $(\cos\theta + i\sin\theta)^5$, showing at least the real terms. Condone slips in powers of i.
	$\cos 5\theta = \cos^5\theta - 10\cos^3\theta\sin^2\theta$ $+ 5\cos\theta\sin^4\theta$	**A1**	
	$\cos 5\theta = \cos\theta\big((1-\sin^2\theta)^2$ $- 10(1-\sin^2\theta)\sin^2\theta + 5\sin^4\theta\big)$	**M1**	Replaces $\cos^2\theta$ by $1-\sin^2\theta$ in their real part, keeping a factor $\cos\theta$.
	$= \cos\theta\big(1 - 2\sin^2\theta + \sin^4\theta$ $- 10\sin^2\theta + 15\sin^4\theta\big)$	**dM1**	Expands and collects into powers of $\sin\theta$. Condone arithmetic slips.
	$\cos 5\theta \equiv \cos\theta(1 - 12\sin^2\theta + 16\sin^4\theta)$	**A1 AG**	Correct substitution leading without error to the printed identity; expansion and collection may be combined.
		[5]	
12(ii)	$\frac{\cos 5\theta}{\cos\theta} = 1 - 12u + 16u^2$, $u = \sin^2\theta$	**M1**	Divides the identity by $\cos\theta$ and treats the result as a quadratic in $\sin^2\theta$, any letter.
	$0 \leq u < 1$	**B1**	Strict inequality at 1 needed; equivalent statements in $\sin\theta$ accepted.
	$u = \frac{12}{2\times 16}$ $[= \frac{3}{8}]$	**M1**	Finds the turning point of their quadratic by differentiation, completing the square or $-\frac{b}{2a}$. Condone sign slips.
	$1 - 12\left(\frac{3}{8}\right) + 16\left(\frac{3}{8}\right)^2 = -\frac{5}{4}$	**A1**	$-\frac{5}{4}$ oe.
	$u \to 1 \Rightarrow 1 - 12 + 16 = \ldots$	**M1**	Evaluates their quadratic at, or as u tends to, $u = 1$.
	5, not attained: $u = 1 \Rightarrow \cos\theta = 0$	**A1**	5 with the reason it is not attained. Solving $f(u) = 5$ is not required.
	$-\frac{5}{4} \leq \frac{\cos 5\theta}{\cos\theta} < 5$	**A1**	Correct inequality at both ends; $\left[-\frac{5}{4}, 5\right)$ oe accepted.
		[7]	
	Total for Question 12	**12**	

Question	Scheme	Marks	Notes
13(a)	$\cos 5\theta + i\sin 5\theta = (c+is)^5$ $= c^5 + 5ic^4s - 10c^3s^2 - 10ic^2s^3$ $+ 5cs^4 + is^5$	**M1**	Equates $\cos 5\theta + i\sin 5\theta$ to a binomial expansion of $(c+is)^5$ showing at least the real terms. Condone sign slips from powers of i.
	$\cos 5\theta = c^5 - 10c^3s^2 + 5cs^4$	**A1**	
	$= c^5 - 10c^3(1-c^2) + 5c(1-c^2)^2$	**M1**	Replaces s^2 by $1-c^2$ throughout their real part, including s^4 by $(1-c^2)^2$.
	$= c^5 - 10c^3 + 10c^5 + 5c - 10c^3 + 5c^5$	**dM1**	Multiplies out all brackets and collects like terms to a polynomial in $\cos\theta$ only. Condone sign slips.
	$\cos 5\theta = 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta$	**A1 AG**	cso: correct substituted real part leading without error to the printed identity; combined steps accepted.
		[5]	
13(b)	$x = \cos\theta$: $4(16x^5 - 20x^3 + 5x) + 1 = 4\cos 5\theta + 1 = 0$	**M1**	Substitutes $x = \cos\theta$ and uses part (a) to reach an equation in $\cos 5\theta$. Roots found without the identity score nothing in this part.
	$\cos 5\theta = -\frac{1}{4}$	**A1**	
	$5\theta = 104.48°,\ 255.52°$ $\Rightarrow \theta = 20.90°,\ 51.10° \Rightarrow x = \cos\theta = \ldots$	**M1**	Finds arccos of their $\cos 5\theta$ and a second solution such as $360° - \alpha$ (or $2\pi - \alpha$), divides by 5 and takes cosines.
	$x = 0.934,\ 0.628$	**A1**	awrt 0.934 and 0.628, with no other positive roots given.
		[4]	
	Total for Question 13	**9**	

Question	Scheme	Marks	Notes
14(a)	$\cos 4\theta + i\sin 4\theta = (\cos\theta + i\sin\theta)^4$ $= \cos^4\theta + 4i\cos^3\theta\sin\theta - 6\cos^2\theta\sin^2\theta$ $- 4i\cos\theta\sin^3\theta + \sin^4\theta$	**M1**	Equates $\cos 4\theta + i\sin 4\theta$ to a binomial expansion of $(\cos\theta + i\sin\theta)^4$ showing at least the three real terms. Condone slips in coefficients or powers of i.
	$\cos 4\theta = \cos^4\theta - 6\sin^2\theta\cos^2\theta + \sin^4\theta$	**A1**	
	$\cos 4\theta = (\sin^2\theta + \cos^2\theta)^2 - 8\sin^2\theta\cos^2\theta$	**M1**	Uses $\sin^2\theta + \cos^2\theta = 1$ to rewrite the fourth powers, e.g. completing the square or substituting $\cos^2\theta = 1 - \sin^2\theta$.
	$\cos 4\theta = 1 - 8\sin^2\theta\cos^2\theta$	**dM1**	Collects to the form $\cos 4\theta = 1 - k\sin^2\theta\cos^2\theta$.
	$\sin^2\theta\cos^2\theta = \frac{1}{8}(1 - \cos 4\theta)$	**A1 AG**	Printed result reached without error from a correct real-part equation; combined steps accepted. Nothing for working back from the result.
		[5]	
14(b)	$e^{\int -2\tan\theta\,\mathrm{d}\theta} = e^{-2\ln\sec\theta} = \ldots$	**M1**	Forms $e^{\int \pm 2\tan\theta\,\mathrm{d}\theta}$ and integrates to the form $e^{k\ln\sec\theta}$ or $e^{k\ln\cos\theta}$.
	$= \cos^2\theta$	**A1**	
	$\frac{\mathrm{d}}{\mathrm{d}\theta}(y\cos^2\theta) = \sin^2\theta\cos^2\theta = \frac{1}{8}(1 - \cos 4\theta)$ $\Rightarrow y\cos^2\theta = \ldots$	**dM1**	Multiplies through by their integrating factor, writes LHS as $\frac{\mathrm{d}}{\mathrm{d}\theta}(y\times$ their factor) and integrates the RHS, using (a) or otherwise, with at least one term correct.
	$y\cos^2\theta = \frac{\theta}{8} - \frac{\sin 4\theta}{32} + c$	**A1**	oe; needs the constant.
	$0 = \frac{\pi}{32} - \frac{\sin\pi}{32} + c \Rightarrow c = \ldots\ [= -\frac{\pi}{32}]$	**M1**	Substitutes $\theta = \frac{\pi}{4}$, $y = 0$ into their general solution containing a constant and solves for it.
	$y = \frac{4\theta - \sin 4\theta - \pi}{32\cos^2\theta}$	**A1**	oe; $y\cos^2\theta = \frac{\theta}{8} - \frac{\sin 4\theta}{32} - \frac{\pi}{32}$ accepted.
		[6]	
Total for Question 14		**11**	

Question	Scheme	Marks	Notes
15(a)	$(z+z^{-1})^6 = 64\cos^6\theta$	**B1**	
	$(z+z^{-1})^6 = z^6+6z^4+15z^2+20$ $\quad +15z^{-2}+6z^{-4}+z^{-6}$	**M1 A1**	**M1** Attempts the binomial expansion: seven terms with binomial coefficients and powers of z. Powers need not be simplified; condone one coefficient slip. **A1** Correct expansion with powers simplified.
	$64\cos^6\theta = (z^6+z^{-6})+6(z^4+z^{-4})+\ldots$ $\quad = 2\cos 6\theta + 12\cos 4\theta + 30\cos 2\theta + 20$	**M1**	Pairs z^n with z^{-n} and replaces by $2\cos n\theta$ for at least two pairs, with their coefficients.
	$a=\frac{1}{32},\ b=\frac{3}{16},\ c=\frac{15}{32},\ d=\frac{5}{16}$	**A1**	All four, oe, stated or in the identity.
		[5]	
15(b)	$\frac{1}{32}\int_0^{\pi/4}(\cos 6\theta + 6\cos 4\theta + 15\cos 2\theta + 10)\,\mathrm{d}\theta$ $= \frac{1}{32}\left[\frac{1}{6}\sin 6\theta + \frac{3}{2}\sin 4\theta + \frac{15}{2}\sin 2\theta + 10\theta\right]_0^{\pi/4}$	**M1 A1**	**M1** Integrates their part (a) identity term by term to the form $p\sin 6\theta + q\sin 4\theta + r\sin 2\theta + s\theta$. Condone sign and coefficient slips. **A1** Correct integral, oe.
	$= \frac{1}{32}\left(-\frac{1}{6}+0+\frac{15}{2}+\frac{5\pi}{2}\right)-0$ $= \frac{11}{48}+\frac{5\pi}{64}$	**dM1** **A1 AG**	**dM1** Substitutes $\theta=\frac{\pi}{4}$ and $\theta=0$ and subtracts; the zero lower limit may be implied. **A1 AG** Correct substituted expression with sine values evaluated, leading without error to the printed result; combined steps accepted.
		[4]	
	Total for Question 15	**9**	

Question	Scheme	Marks	Notes
16(i)(A)	$(\cos\theta + i\sin\theta)^n = (e^{i\theta})^n = e^{in\theta}$	**M1**	Raises both sides of Euler's formula to the power n and writes $(e^{i\theta})^n = e^{in\theta}$.
	$e^{in\theta} = \cos n\theta + i\sin n\theta$	**A1 AG**	Applies Euler's formula with $n\theta$ to reach the printed result. Quoting de Moivre's theorem earns nothing.
		[2]	
16(i)(B)	$e^{-i\theta} = \cos(-\theta) + i\sin(-\theta)$ $= \cos\theta - i\sin\theta$	**M1**	Obtains $e^{-i\theta}$ in terms of $\cos\theta$ and $\sin\theta$, e.g. by replacing θ by $-\theta$, and subtracts the two results. Condone sign slips.
	$e^{i\theta} - e^{-i\theta} = 2i\sin\theta$ $\Rightarrow \sin\theta = \frac{1}{2i}(e^{i\theta} - e^{-i\theta})$	**A1 AG**	Correct $e^{-i\theta} = \cos\theta - i\sin\theta$, with $\cos(-\theta) = \cos\theta$ and $\sin(-\theta) = -\sin\theta$ used, leading without error to the printed result.
		[2]	
16(ii)	$\cos 8\theta + i\sin 8\theta = (\cos\theta + i\sin\theta)^8$ $= \cos^8\theta + 8i\cos^7\theta\sin\theta$ $- 28\cos^6\theta\sin^2\theta + \ldots$	**M1**	Uses de Moivre with $n = 8$ and expands $(\cos\theta + i\sin\theta)^8$ binomially, at least the real terms. Condone coefficient or sign slips.
	$\cos 8\theta = \cos^8\theta - 28\cos^6\theta\sin^2\theta$ $+ 70\cos^4\theta\sin^4\theta - 28\cos^2\theta\sin^6\theta$ $+ \sin^8\theta$	**A1**	
	$\cos 8\theta = (1 - \sin^2\theta)^4$ $- 28(1 - \sin^2\theta)^3\sin^2\theta + \ldots$	**M1**	Replaces every $\cos^2\theta$ in their real part by $1 - \sin^2\theta$.
	$\cos 8\theta = (1 - 4\sin^2\theta + 6\sin^4\theta - \ldots)$ $- 28(\sin^2\theta - 3\sin^4\theta + \ldots) + \ldots$	**dM1**	Expands all the brackets and collects like powers of $\sin\theta$. Condone sign and arithmetic slips.
	$a = 128,\ b = -256,\ c = 160,\ d = -32,\ e = 1$	**A1 A1**	**A1** At least three of the five constants correct. **A1** All five correct; the identity with these coefficients is accepted.
		[6]	
16(iii)	$\sin^8\theta = \frac{1}{(2i)^8}(e^{i\theta} - e^{-i\theta})^8$ $= \frac{1}{256}(e^{i\theta} - e^{-i\theta})^8$	**B1**	Needs $(2i)^8$ evaluated as 256.

Question 16 continued

Question	Scheme	Marks	Notes
	$= \frac{1}{256}(e^{8i\theta} - 8e^{6i\theta} + 28e^{4i\theta} - 56e^{2i\theta} + 70$ $- 56e^{-2i\theta} + 28e^{-4i\theta} - 8e^{-6i\theta} + e^{-8i\theta})$	**M1 A1**	**M1** Expands the eighth power binomially with coefficients from row 8 of Pascal's triangle, combining powers of $e^{i\theta}$. Accept $z = e^{i\theta}$ notation. Condone sign slips. **A1** All nine terms correct.
	$\sin^8\theta = \frac{1}{256}(2\cos 8\theta - 16\cos 6\theta + 56\cos 4\theta$ $- 112\cos 2\theta + 70)$	**M1**	Pairs $e^{ik\theta}$ with $e^{-ik\theta}$ as $2\cos k\theta$ for at least two values of k.
	$P = \frac{1}{128},\ Q = -\frac{1}{16},\ R = \frac{7}{32},$ $S = -\frac{7}{16},\ T = \frac{35}{128}$	**A1 A1**	**A1** At least three of the five constants correct. **A1** All five correct, oe.
		[6]	
16(iv)	$\sin^8\frac{\pi}{8} = \frac{1}{128}\left(-1 - 8\left(-\frac{\sqrt{2}}{2}\right) + 28(0)\right.$ $\left. -56\left(\frac{\sqrt{2}}{2}\right) + 35\right)$	**M1**	Substitutes $\theta = \frac{\pi}{8}$ into their identity of the form in (iii) and uses exact cosine values. Decimal cosines do not score.
	$\sin^8\frac{\pi}{8} = \frac{34-24\sqrt{2}}{128} = \frac{17-12\sqrt{2}}{64}$	**A1**	cso, from correct constants in (iii).
	$0 < \frac{\pi}{8} < \frac{\pi}{2} \Rightarrow \sin\frac{\pi}{8} > 0$ $\Rightarrow \sin\frac{\pi}{8} = \left(\frac{17-12\sqrt{2}}{64}\right)^{\frac{1}{8}}$	**A1 AG**	Needs a reason for taking the positive real eighth root, such as $\sin\frac{\pi}{8} > 0$ or $\frac{\pi}{8}$ acute.
		[3]	
	Total for Question 16	**19**	

Question	Scheme	Marks	Notes
17(a)	$(z-z^{-1})^4(z+z^{-1})^2 = (2i\sin\theta)^4(2\cos\theta)^2$ $= 64\sin^4\theta\cos^2\theta$	**B1**	
	$(z-z^{-1})^4(z+z^{-1})^2 = (z^2-2+z^{-2})(z^4-2+z^{-4})$ $= \ldots$	**M1**	Attempts the full expansion of the product into powers of z, in any order. Condone sign and coefficient slips.
	$= z^6 - 2z^4 - z^2 + 4 - z^{-2} - 2z^{-4} + z^{-6}$	**A1**	Correct seven-term expansion, oe.
	$= (z^6 + z^{-6}) - 2(z^4 + z^{-4}) - (z^2 + z^{-2}) + 4$ $= 2\cos 6\theta - 4\cos 4\theta - 2\cos 2\theta + 4$	**M1**	Pairs z^n with z^{-n} and replaces each pair by $2\cos n\theta$.
	$64\sin^4\theta\cos^2\theta = 2\cos 6\theta - 4\cos 4\theta - 2\cos 2\theta + 4$ $\Rightarrow \sin^4\theta\cos^2\theta \equiv \frac{1}{32}(\cos 6\theta - 2\cos 4\theta - \cos 2\theta + 2)$	**A1 AG**	Correct expansion equated to $64\sin^4\theta\cos^2\theta$, leading without error to the printed identity.
	Alternative: Double-angle formulae in $\cos 2\theta$		
	$\sin^4\theta\cos^2\theta = \left(\frac{1-\cos 2\theta}{2}\right)^2\left(\frac{1+\cos 2\theta}{2}\right) = \ldots$	**M1**	Uses $\sin^2\theta = \frac{1}{2}(1-\cos 2\theta)$, $\cos^2\theta = \frac{1}{2}(1+\cos 2\theta)$ and expands to a polynomial in $\cos 2\theta$. Condone sign slips.
	$= \frac{1}{8}(1 - \cos 2\theta - \cos^2 2\theta + \cos^3 2\theta)$	**A1**	
	$(\cos 2\theta + i\sin 2\theta)^3 = \cos 6\theta + i\sin 6\theta$ $\Rightarrow \cos 6\theta = \cos^3 2\theta - 3\cos 2\theta\sin^2 2\theta = \ldots$	**M1**	Expands $(\cos 2\theta + i\sin 2\theta)^3$ and equates real parts to express $\cos 6\theta$ in powers of $\cos 2\theta$; $\cos 4\theta$ may come from a double-angle formula. Not for quoting the triple-angle formula.
	$\cos^2 2\theta = \frac{1}{2}(1 + \cos 4\theta)$ $\cos^3 2\theta = \frac{1}{4}(\cos 6\theta + 3\cos 2\theta)$	**A1**	Both, oe.
	$\sin^4\theta\cos^2\theta =$ $\frac{1}{8}\left(1 - \cos 2\theta - \frac{1+\cos 4\theta}{2} + \frac{\cos 6\theta + 3\cos 2\theta}{4}\right)$ $\equiv \frac{1}{32}(\cos 6\theta - 2\cos 4\theta - \cos 2\theta + 2)$	**A1 AG**	Correct substitution into the correct cubic, leading without error to the printed identity.
		[5]	
17(b)	$\sin^2\theta\cos^4\theta = \sin^4\alpha\cos^2\alpha$ $= \frac{1}{32}(\cos 6\alpha - 2\cos 4\alpha - \cos 2\alpha + 2)$	**M1**	Replaces $\sin\theta$ by $\cos\alpha$ and $\cos\theta$ by $\sin\alpha$ and applies the identity of (a) in α.
	$\cos 2\alpha = -\cos 2\theta,\ \cos 4\alpha = \cos 4\theta,\ \cos 6\alpha = -\cos 6\theta$	**M1**	Uses $n\alpha = \frac{n\pi}{2} - n\theta$ to write all three cosines in terms of θ. Condone sign slips.
	$\sin^2\theta\cos^4\theta \equiv \frac{1}{32}(-\cos 6\theta - 2\cos 4\theta + \cos 2\theta + 2)$	**A1**	oe.
		[3]	

Question 17 continued

Question	Scheme	Marks	Notes
17(c)	$\frac{1}{32}(\cos 6\theta - 2\cos 4\theta - \cos 2\theta + 2)$ $+ \frac{2}{32}(\text{“} - \cos 6\theta - 2\cos 4\theta + \cos 2\theta + 2\text{”})$	**M1**	Adds the identity from (a) to twice their identity from (b), of the form $\frac{1}{32}$(cosines of $2\theta, 4\theta, 6\theta$ and a constant).
	$= \frac{1}{32}(-\cos 6\theta - 6\cos 4\theta + \cos 2\theta + 6)$	**A1**	
	$\frac{1}{32}\left[-\frac{1}{6}\sin 6\theta - \frac{3}{2}\sin 4\theta + \frac{1}{2}\sin 2\theta + 6\theta\right]_0^a$	**M1**	Integrates at least one cosine term to the form $k\sin n\theta$ and substitutes the limits 0 and a.
	$\frac{1}{32}\left(6a + \frac{1}{2}\sin 2a - \frac{3}{2}\sin 4a - \frac{1}{6}\sin 6a\right)$ $= \frac{\pi}{32} - \frac{\sqrt{3}}{64}$	**A1**	Correct expression in a, oe, equated to the given value.
	$a = \frac{\pi}{6}$	**A1**	Exact; may be found by trial and checked by substitution. Uniqueness need not be shown.
		[5]	
	Total for Question 17	**13**	

Question	Scheme	Marks	Notes
18(a)	$z = \cos\theta + i\sin\theta \Rightarrow (z+z^{-1})^5 = 32\cos^5\theta$	**B1**	
	$(z+z^{-1})^5 = z^5 + 5z^3 + 10z + 10z^{-1} + 5z^{-3} + z^{-5}$	**M1 A1**	**M1** Expands to six terms in $z^5, z^3, \ldots, z^{-5}$ with binomial coefficients. Condone coefficient slips. **A1** Correct expansion.
	$= 2\cos 5\theta + 10\cos 3\theta + 20\cos\theta$	**M1**	Pairs terms as $z^n + z^{-n}$ and replaces each pair by $2\cos n\theta$.
	$A = \frac{1}{16},\ B = \frac{5}{16},\ C = \frac{5}{8}$	**A1**	Values stated, or the identity written with them.
	Alternative: Multiple-angle polynomials		
	$\cos 5\theta = \mathrm{Re}(\cos\theta + i\sin\theta)^5$ $= \cos^5\theta - 10\cos^3\theta\sin^2\theta + 5\cos\theta\sin^4\theta$ $\cos 3\theta = \cos^3\theta - 3\cos\theta\sin^2\theta$	**M1**	Expands $(\cos\theta + i\sin\theta)^n$ and equates real parts, for $n = 5$ at least; $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$ may be quoted. Condone coefficient slips.
	$\cos 5\theta = 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta$ $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$	**A1**	Both correct, using $\sin^2\theta = 1 - \cos^2\theta$.
	$16A = 1,\ -20A + 4B = 0,\ 5A - 3B + C = 0$	**M1**	Substitutes their polynomials into the identity and equates coefficients of at least two powers of $\cos\theta$.
	$A = \frac{1}{16},\ B = \frac{5}{16},\ C = \frac{5}{8}$	**dM1 A1**	**dM1** Solves their equations for A, B and C. **A1** All three correct.
		[5]	
18(b)	$\theta = \arccos(x^{1/6})$ $f'(x) = -(15\cos 5\theta + 75\cos 3\theta + 150\cos\theta)\frac{\mathrm{d}\theta}{\mathrm{d}x}$	**M1 A1**	**M1** Differentiates each $\sin k\theta$ to $\pm k\cos k\theta$ times $\frac{\mathrm{d}\theta}{\mathrm{d}x}$ or an inner derivative. Condone coefficient slips. **A1** Correct outer derivatives; the inner derivative may be unevaluated or incorrect.
	$\frac{\mathrm{d}\theta}{\mathrm{d}x} = -\frac{1}{\sqrt{1-(x^{1/6})^2}} \times \frac{1}{6}x^{-5/6}$	**M1**	Uses $\frac{\mathrm{d}}{\mathrm{d}u}\arccos u = \pm\frac{1}{\sqrt{1-u^2}}$ with $\frac{\mathrm{d}u}{\mathrm{d}x} = kx^{-5/6}$, or differentiates $\cos\theta = x^{1/6}$ implicitly.
	$= -\frac{x^{-5/6}}{6\sqrt{1-x^{1/3}}}$	**A1**	oe, in terms of x.
	$15(\cos 5\theta + 5\cos 3\theta + 10\cos\theta) = 15 \times 16\cos^5\theta$	**M1**	Uses part (a) to replace the cosine sum by a multiple of $\cos^5\theta$.

Question 18 continued

Question	Scheme	Marks	Notes
	$f'(x) = 240x^{5/6} \times \frac{x^{-5/6}}{6\sqrt{1-x^{1/3}}}$ $= \frac{40}{\sqrt{1-x^{1/3}}}$	**A1 AG**	Correct substituted expression with $\cos^5\theta = x^{5/6}$, leading without error to the printed result.
		[6]	
18(c)	Area $= \frac{1}{40}\int_0^1 f'(x)\,\mathrm{d}x = \frac{1}{40}\left[f(x)\right]_0^1$	**M1**	Uses part (b) to write the area as $\frac{1}{40}$ of f between 0 and 1. Limit notation at 1 need not be shown.
	$x \to 1 \Rightarrow f(x) \to 0$; $f(0) = -3 + 25 - 150 = -128$ $\Rightarrow \frac{1}{40}\big(0 - (-128)\big)$	**dM1**	Uses $\arccos 1 = 0$ and $\arccos 0 = \frac{\pi}{2}$ to evaluate f at both limits and subtracts in the correct order. Condone one sign slip in $f(0)$.
	$\frac{16}{5}$	**A1**	cso; 3.2 accepted.
		[3]	
	Total for Question 18	**14**	